

EXPERIMENTAL AND THEORETICAL INVESTIGATIONS ON THE PHYSICS OF FAST REACTORS*

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1. Study of a Model of a Reactor of the BN-350 Type on the BFS Rig

At the 3rd Geneva Conference⁽¹⁾ fundamental results were given of a study of the physics of fast reactors on the BFS rig for several critical assemblies.

After the conference, investigations of a model of the BOR reactor (BFS-13 and BFS-14 assemblies) were carried out on the BFS rig, but attention was principally devoted to experiments on simulation of the BN-350 type reactor⁽²⁾ and an analysis of these experiments. The main results of the work are given below.

Description of the Critical Assemblies

Several versions of critical assemblies were investigated for the purpose of studying the physical characteristics of the reactor—a model of BN-350 with the actual arrangement and worth of the real control rods (BFS-15-1 and BFS-15-3 assemblies). Then all the control rod simulators were replaced by fuel elements. In this assembly there were no inhomogeneities due to the control and safety rods, which made it easier to compare the calculations and experiments—this was the pure BFS-16 assembly.

In all the assemblies the core was made up of two types of element with

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TABLE 1. COMPOSITION OF DIFFERENT ZONES OF CRITICAL ASSEMBLY ($\times 10^{-22}$ nucl/cm³)

	Core		End blankets		Interlayer between core and end blankets		Lateral blanket
	Low enrichment	High enrichment	Top	Bottom	Top	Bottom	
U-235	0.1406	0.2185	—	—	—	—	—
U-238	0.661	0.5788	0.9826	0.9826	—	—	1.17
Oxygen	1.578	1.611	1.974	1.974	—	—	2.35
Aluminium	0.9315	0.938	1.256	1.256	1.5	1.73	0.593
Steel $\rho = 8.47$	2.053	2.05	0.583	0.583	3.663	3.686	1.364

The low-enrichment zone has a radius of 55.35 cm.

Height of core 106 cm.

Thickness of interlayers:

top 9 cm.

bottom 6.5 cm.

top 60 cm.

Thickness of end blankets:

bottom 24 cm.

TABLE 2. COMPOSITION OF SAFETY AND CONTROL ELEMENT SIMULATORS ($\times 10^{-22}$ nucl/cm³)

Element	Burnup compensator (KP)		Automatic control rod (AR)		Safety (AZ) and temperature compensator (TK)	
	In	Out	In	Out	In	Out
U-235	0.0893	0.0024	—	—	—	—
U-238	0.430	0.350	—	—	—	—
Oxygen	1.353	0.70	—	—	—	—
Aluminium	0.950	1.44	1.24	1.09	0.295	1.62
Steel	2.65	2.76	0.736	3.44	1.10	3.25
Boron-10	—	—	0.685	—	0.868	—
Boron-11	—	—	0.171	—	3.40	—
Carbon	—	—	0.214	—	1.50	—
Number of rods and their distance from the centre	6 (KP-1—17.8 cm, KP-2—28.0 cm, KP-3—25.1 cm, KP-4—17.8 cm, KP-5—25.1 cm, KP-6—25.1 cm)		2 (AR-1—10.2 cm, AR-2—10.2 cm)		3 AZ+1 TK (AZ-1—26.9 cm, AZ-2—25.1 cm, AZ-3—29.4 cm, TK—26.9 cm)	

TABLE 3. CRITICAL PARAMETERS OF ASSEMBLIES

		15-1	15-2	15-3	16-1	16-2
Radius of core, cm		70.1	72.25	71.05	65.7	66.5
Critical mass U-235		1055*	1118*	1073†	919	946
Theoretical multiplication factor K_{eff}	One-dimensional calculation diffusion approximation	0.9974	—	0.9865	0.9945	0.9947
	One-dimensional calculation P_3 approximation	1.0058	—	0.9974	1.0002	1.0022
	Two-dimensional calculation diffusion approximation	—	—	1.0077	1.0019	1.0025
Theoretical critical masses with U-235 loaded into the edge of the core	One-dimensional diffusion calculation	1066	—	1135	941	967
	One-dimensional calculation in P_3 approximation	1031	—	1085	919	937
	Two-dimensional diffusion calculation	—	—	1037	912	936

* The critical masses and Ra are given for thin blankets; in the critical masses allowance is made for the U-235 in the compensators.
† Taking into account the U-235 in the compensators.

TABLE 4. GROUP ENERGY INTERVALS

E_{low} , eV	10,000	4650	2150	1000	465	215	100	46.5	21.5	10	4.65	2.15	1	0.465	0.215	Thermal
Group No. in 26-group system	1-11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26
Group No. in 18-group system	1-11	12	12	13	13	14	14	15	15	15	16	16	17	17	17	18

different U-235 contents, simulating two enrichment zones. The compositions of the different zones are shown in Table 1. All changes made in the reactor

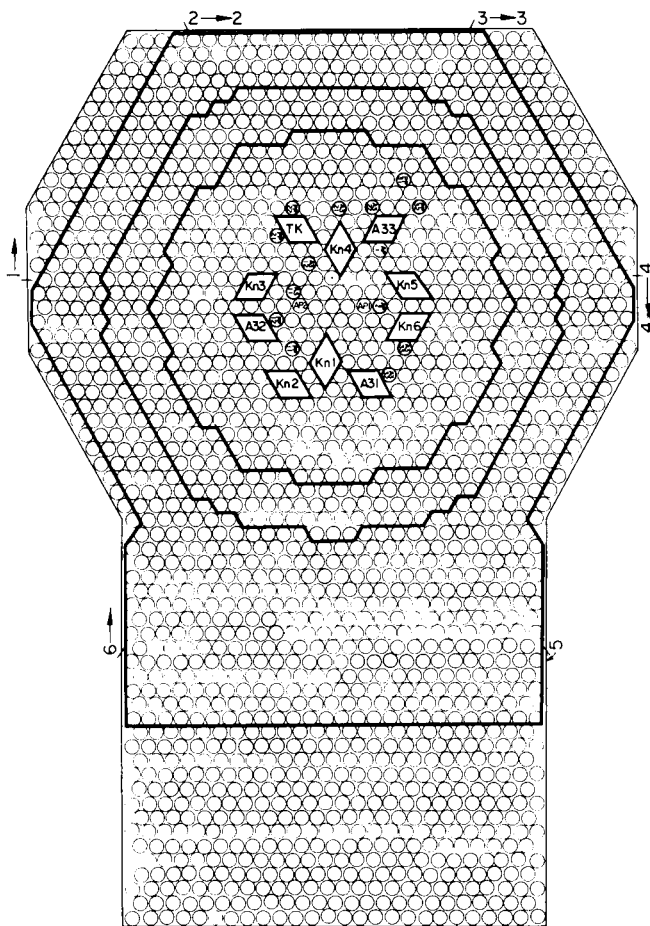


FIG. 1. Diagram of the BFS-15-3 assembly.

TK = temperature compensator.

A3 = boron safety element.

K = burnup compensator.

AP = automatic control element.

were balanced out by adding or removing some fuel elements at the peripheral high-enrichment zone. The radius of the low-enrichment zone was not changed.

During the experiments and when analysing them, different positions of the control and safety rod simulators were considered. In the BFS-15 assembly there were simulators for three boron safety rods (AZ), one boron temperature compensator (TK), six burnup compensators (KP), and two automatic

control rods (AR). Table 2 gives the compositions of the safety and control rod simulators in different positions and at different distances from the centre (Fig. 1).

The main results given below were obtained on the following critical assemblies:

- BFS-15-1. Condition with minimum critical mass—all compensators in the "in" position and the boron elements "out".
- BFS-15-2. Compensators and boron rods in the "out" position.
- BFS-15-3. The same as for BFS-15-2 but one control rod "in".
- BFS-15-4. The same as for BFS-15-3 but one boron temperature compensator was in the "in" position.
- BFS-16-1. "Pure" assembly without simulators for safety and control rods.
- BFS-16-2.* As above, with central boron control rod.

The first two assemblies had a thin lateral blanket (about 20 cm). In all the rest the blanket was increased in one direction to 60 cm, the effect of increasing the thickness was determined and extrapolation to this thickness was carried out for the whole lateral blanket.

Critical Parameters

The experimental and theoretical critical characteristics of several assemblies are given in Table 3.

The main calculations were carried out in a one-dimensional 18-group diffusion approximation, but for some calculations a one-dimensional P_3 and two-dimensional 18-group diffusion approximation were used. The 18-group constants were obtained by enlarging the energy intervals of the 26-group system⁽³⁾ in the range below 10 keV. The 26- and 18-group breakdown is shown in Table 4. The cross-sections used took resonance shielding into account. The elastic slowing-down cross-sections took into account the form of the neutron spectrum.

In the calculations of assemblies 15-1–15-3 homogenization of the composition was carried out in the region where the control and safety element simulators were arranged. The radius of the homogenized region was 35.2 cm. The error in this homogenization, assessed by perturbation theory, was no greater than 0.1% $\Delta k/k$. The centrally placed boron control rods (AR) were separated into an individual zone with a radius of 2.7 cm.

To obtain the theoretical values of the critical masses (Table 3) from the value K_{eff} , use was made of an experimental relationship between the reactivity and the number of rods loaded. Thus for version 15-3, the effectiveness of one peripheral fuel rod is $3.2 \times 10^{-4} \Delta k/k$.

* In the following the assemblies will sometimes be referred to as 15-1, 16-1, etc., for the sake of brevity.

The K_{eff} values in Table 3 do not take into account the heterogeneous structure of the assemblies. Some results of a study of the effects of heterogeneity on the BFS rig are given below.

Measurement of the Cross-section Ratios in the Centre of the Assemblies and Distribution of the Processes

The cross-section ratios were measured in assemblies 15-3 and 16-1. The ratios of the average fission cross-sections of U-238, Pu-239, U-233, Th-232 and Np-237 to the average fission cross-section of U-235 were determined.

TABLE 5. CROSS-SECTION RATIO IN THE CENTRE

	15-3		16-1	
	Experiment	Calculation	Experiment	Calculation
σ_f^8/σ_f^5	0.0231 ± 0.0004	0.0225	0.0239 ± 0.0004	0.0258
σ_f^9/σ_f^5	0.97 ± 0.02	0.983	0.99 ± 0.02	0.989
σ_f^3/σ_f^5	1.47 ± 0.03	1.44	1.48 ± 0.03	1.44
σ_f^2/σ_f^5	—	—	0.0051 ± 0.0005	0.005
σ_f^7/σ_f^5	—	—	0.161 ± 0.009	0.183
σ_c^8/σ_f^5	0.139 ± 0.003	0.140	0.154 ± 0.006	0.139
$\sigma_c^{\text{Cu}}/\sigma_f^5$	—	—	0.0318 ± 0.003	0.0304
$\sigma_c^{\text{Na}}/\sigma_f^5$	—	—	0.00132 ± 0.00006	0.0014
$\sigma_c^{\text{Au}}/\sigma_f^5$	—	—	0.379 ± 0.051	0.334
$\sigma_c^{\text{In}}/\sigma_f^5$	—	—	0.251 ± 0.034	0.214
σ_c^1/σ_f^5	—	—	0.263 ± 0.039	0.216

For these measurements small chambers 8 mm in diameter with a stainless steel wall 0.2 mm thick were used. To check the effect of the wall thickness on the chamber readings with threshold isotopes, special experiments were carried out with different wall thicknesses, showing that with a wall thickness of 0.2 mm the effect of inelastic scattering can be ignored.

Special experiments were undertaken to determine the effect on the chamber readings of electrical cables, insulators and other structural elements (irradiation of U-235 and natural uranium foils in models of chambers and without them). These experiments showed that there was no such effect.

In addition to the chamber measurements, activation measurements of the neutron capture cross-sections were carried out in the centre of the

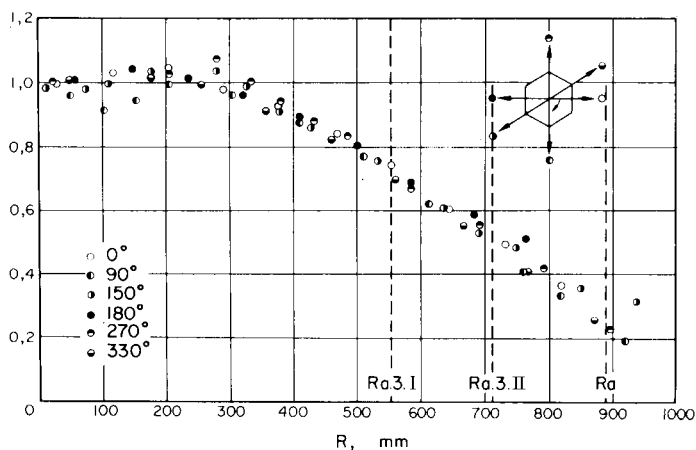


FIG. 2. Experimental U-235 fission distribution in different directions in the 15-3 assembly.

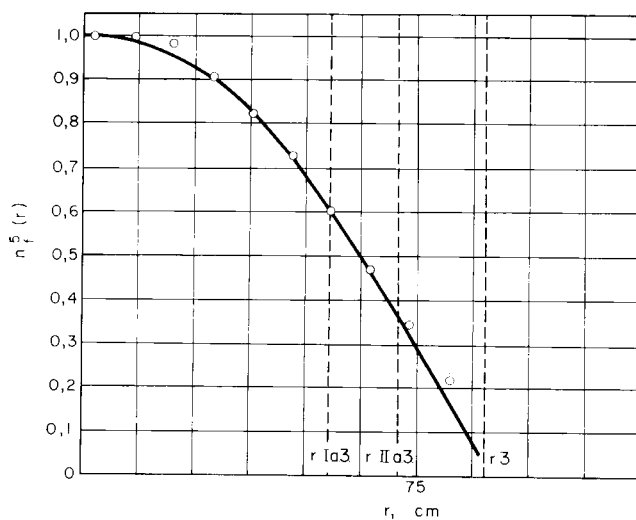


FIG. 3. Theoretical and experimental radial distribution of U-235 fissions in the 15-1 assembly. Continuous line = calculation. Circles = experiment.

assemblies for U-238, Cu-63, Na, Au, In, I (with respect to the fission cross-section of U-235).

In all the assemblies, measurements were made with greater or lesser detail of the distribution of the different processes radially and axially. The most detailed measurements were made on the 15-3 assembly with fission chambers

with U-235 and U-238, and the spatial distribution of captures in U-238 was also determined (Figs. 2, 3 and 4).

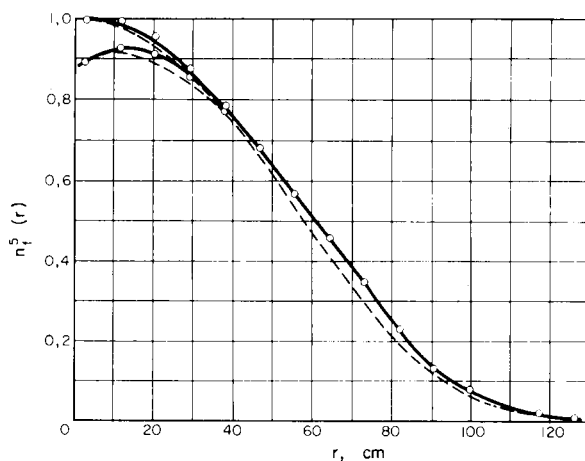


FIG. 4. Effect of boron control rod on the radial distribution of U-235 fissions (16-1 and 16-2 assemblies). Continuous line = experiment. Broken line = calculation.

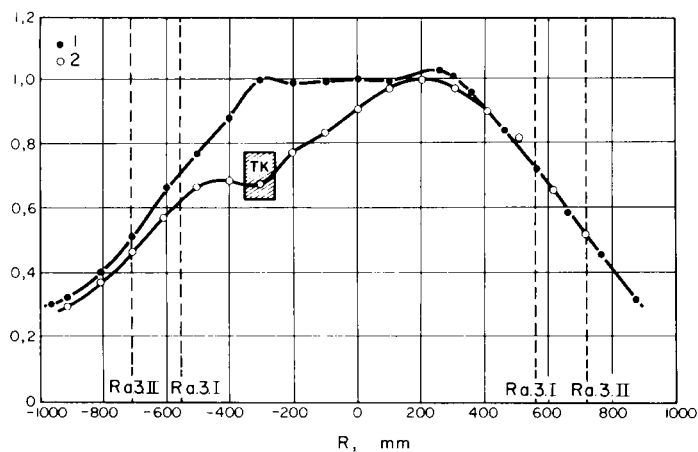


FIG. 5. U-235 fission distribution for different positions of the boron temperature compensator (experiment, assembly 15-4).

The results of the measurements and the calculations of the cross-section ratios are shown in Table 5 (Fig. 5).

A theoretical assessment of the effect of a boron compensator on the U^{235} distribution was made for a centrally placed rod. The relative reduction in the flux near the rod in this calculation was very close to the experimental value.

On the basis of detailed experimental distributions of U-235 fissions and captures in U-238 in the 15-3 assembly, the conversion coefficient

$$KK = \frac{\int_v N_{c8}(r_1 z) dv}{(1 + \bar{\alpha}) \int_v N_{f5}(r_1 z) dv}$$

was determined; the value $\bar{\alpha}$ was taken from the calculations.

This conversion coefficient was 1.05 ± 0.03 . The theoretical value of the conversion coefficient for this assembly, obtained from one-dimensional calculations, is 1.035.

Investigation of Perturbations caused by Different Materials

On the BFS-15-3 and BFS-16-1 assemblies a study was made of the effectiveness of different materials placed in the centre of the core. The measurements were made by means of the reactor period. The perturbations were determined with the specimen in question placed in a prepared cavity in the central channel which matched the dimensions of the specimen. The results of these measurements are given in Table 6. This table includes the theoretical values of the effectiveness obtained on the basis of first-order multi-group perturbation theory.

The experimental and theoretical data for a number of materials (U-235, U-238, Ta, Re, B₄C) differ by about 10%; for the other specimens the difference is much greater. In the calculation of the perturbations caused by such materials as U-238, Ta, Re, Fe, Nb, Mo, and W, an important part is played by the fact that allowance is made for the resonance shielding of the cross-sections of the specimens in view of their fairly large dimensions. The differences existing in the theoretical and experimental effectiveness values may be due to the inaccuracies of the multi-group cross-sections and the inaccuracy of the calculation of the neutron spectra and importance. The latter fact is particularly important for materials whose reactivity is determined to a great extent by scattering effects (Fe, C, U-238).

The differences between the real neutron spectra and importance and the theoretical ones may be caused by various inhomogeneities, in particular, by the location of safety and control element simulators near the specimen. Such inhomogeneities were not taken into account in the calculation of the 15-3 assembly and it is possible that it is precisely this which caused the discrepancy in the sign of the theoretical and experimental effectiveness of carbon in the 15-3 assembly.

The perturbation due to carbon is determined only by the slowing down of the neutrons, i.e. the variation in the importance of the neutrons slowed down in the carbon. This variation is very sensitive to the composition of the zone, as a result of which a homogeneous theoretical model may greatly alter the result.

TABLE 6. PERTURBATIONS DUE TO SPECIMENS PLACED IN THE CENTRE

Material	Weight of specimen (g)	Effectiveness $\Delta K/K/\text{kg} \times 10^3$					
		Assembly 15-3			Assembly 16-1		
		Exper.	Theoret.	Exper. Theoret.	Exper.	Theoret.	Exper. Theoret.
U-235	740.5	+0.91 $\pm$ 0.05	+0.99	0.92	+1.34 $\pm$ 0.02	+1.48	0.905
(90% enrich.)	296.2*						
U-238	3220	-0.068 $\pm$ 0.020	-0.076	0.895	-0.101 $\pm$ 0.001	-0.11	0.92
Ta	1160	-0.264 $\pm$ 0.015	-0.279	0.945	-0.375 $\pm$ 0.020	-0.393	0.995
Re	362	-0.447 $\pm$ 0.022	-0.502	0.89	-0.668 $\pm$ 0.033	-0.710	0.94
Nb	1287	-0.148 $\pm$ 0.009	-	-	-0.235 $\pm$ 0.005	-0.313	0.75
	642.2*						
Fe	1332	-0.024 $\pm$ 0.003	-0.023	1.04	-0.026 $\pm$ 0.001	-0.032	0.81
Mo	813.2	-	-	-	-0.162 $\pm$ 0.001	-0.180	0.9
W	1145.4	-	-	-	-0.156 $\pm$ 0.001	-0.20	0.78
C	277	+0.069 $\pm$ 0.007	-0.061	-	+0.191 $\pm$ 0.012	+0.019	$\sim$ 10
B ₄ C	137.7*						
	122.8	-2.88 $\pm$ 0.10	-2.87	$\sim$ 1.0	-4.61 $\pm$ 0.01	-4.2	1.1
	153.3*						

* Weight of specimens in assembly 16-1.

In the 16-1 assembly which has no inhomogeneities, the sign of the effect for carbon agrees, although the difference in the absolute value of the effect between theory and experiment is considerable.

A fairly large specimen, especially such a strong absorber as boron carbide, makes it necessary to take into account the effect of self-shielding, and

TABLE 7. EFFECT OF SELF-SHIELDING OF BORON

Quantity B_4C	1 lump (30 g)	2 lumps	3 lumps	5 lumps
Effectiveness $\Delta K/K \cdot 10^{-3}/kg$	-5.00	-4.86 ± 0.07	-4.75 ± 0.04	-4.61 ± 0.01

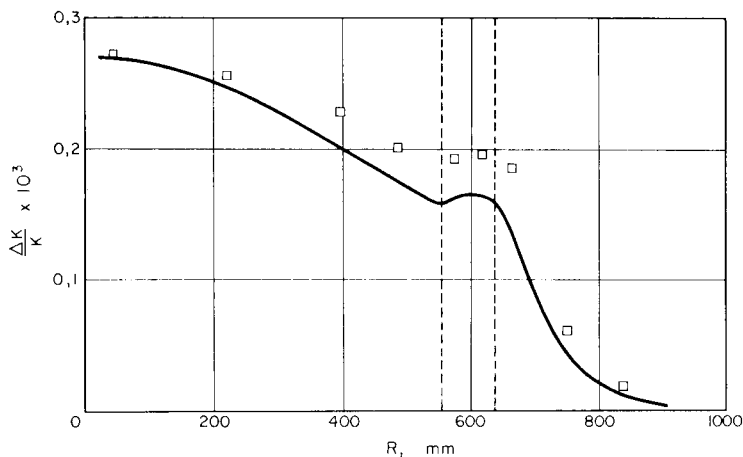


FIG. 6. Experimental and theoretical distribution of the effectiveness of a sodium tube along the radius of the BFS-16 assembly Squares = experiment. Continuous line = calculation).

therefore the appropriate corrections must be made to the perturbation theory calculations. This effect has been studied experimentally for boron carbide specimens on the BFS-16 assembly—Table 7.

As can be seen from the table, the self-shielding effect of boron may amount to 10% in $\Delta K/K$ depending on the weight of the specimen.

On the BFS-16 assembly the radial distribution of the effectiveness of sodium specimens in the reactor was investigated. The experiment was carried out as follows. A channel with only end reflectors and an air cavity up the core was replaced by an identical channel with sodium-filled aluminium containers inside it. The effectiveness was measured by means of the reactor period (Fig. 6).

In assessing this experiment it is necessary to take into account the increase in the neutron leakage both for the empty channel and the channel with sodium. The correction was determined by G. Ya. Rumyantsev's method,^(4, 5) using the P-2 approximation for the calculation of the effective diffusion coefficient of the empty channel.

The following methodical experiment was carried out on the BFS-16 assembly with the aim of studying the interference of strong absorber rods. Four tubes filled with boron carbide through the core (natural enrichment) were placed in the centre of the assembly. The assembly was brought to the critical state by adding fuel to the edge of the core.

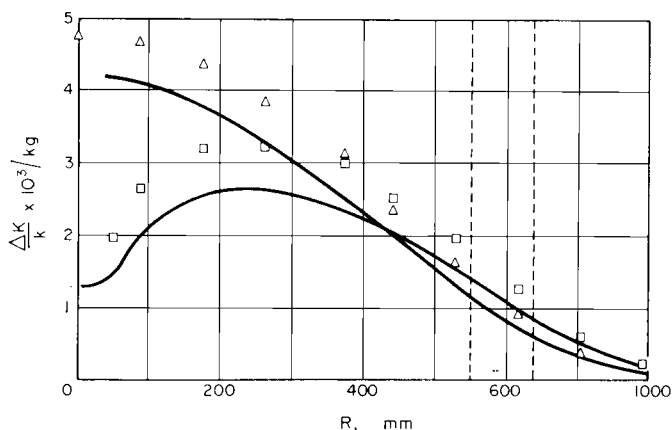


FIG. 7. Experimental and theoretical distribution of the effectiveness of a specimen of boron carbide (natural enrichment) along the radius of the BFS-16 assembly.

□ = experiment in unperturbed system.

△ = experiment with boron rod in the centre of the assembly line.

— = calculation.

In this condition measurements were made of the radial distribution of the effectiveness of the boron carbide specimen in the reactor in the central cavity. The measurements were carried out with the reactor power going up.

Similar measurements were taken on an unperturbed assembly (16-1) (Fig. 7). The calculations were carried out in multi-group diffusion approximation by perturbation theory.

From an analysis of the experimental and theoretical data it is possible to draw the following conclusions:

- The influence of a single strongly absorbing rod on the effectiveness of a small specimen of B_4C extends quite a long way (eight to ten neutron mean free paths).
- The effectiveness of the specimen near the rod decreases by a factor of about two compared with the unperturbed case.

- (c) The theoretical quantities differ from the experimental ones at each point in accordance with the difference of these quantities in the centre of the assembly (Table 6), which shows that the relative behaviour of the relationships given is described fairly well by the existing methods.

Effectiveness of Control Rods

In assemblies 15-1 and 15-3 a study was made of the effectiveness of the simulators for the control and safety rods of the BN-350 reactor. The effectiveness of these elements was calculated by perturbation theory with

TABLE 8. EFFECTIVENESS OF SAFETY AND CONTROL RODS (% $\Delta K/K$)

	Assembly 15-1		Assembly 15-3	
	Experiment	Calculation	Experiment	Calculation
AP ₁	-0.299	-0.328	-0.248	-0.285
AP ₂	-0.295	-0.328	-0.247	-0.285
AP ₁ with AP ₂ out			-0.246	
AP ₂ with AP ₁ out			-0.244	
A3 ₁	-1.5	-1.37	-1.39	-1.24
A3 ₂	-1.47	-1.41	-1.27	-1.25
A3 ₃	-1.38	-1.33	-1.24	-1.22
A3 ₁ + A3 ₂ + A3 ₃	-4.21	—	—	—
K ₁	0.243	0.252	0.202	0.212
K ₂	0.222	0.228	0.189	0.204
K ₃	0.217	0.236	0.185	0.207
K ₄	0.233	0.252	0.192	0.212
K ₅	0.228	0.236	0.186	0.207
K ₆	0.231	0.236	0.190	0.207
K _i	1.32	1.440	—	—

allowance for the flattening of the neutron field near the rods if this effect took place.

The effectiveness values for the different safety and control rod simulators are given in Table 8. Calculation gives a fairly good prediction, but it is possible that this is assisted by mutual compensation of the errors in the determination of the effectiveness of the different components.

The differences in the effectiveness of one and the same rod in the two assemblies are due to the difference in the core size and the different relative flux level and importance at the locations of the elements.

The experiments showed slight interaction of the safety and control rods for the model of the BN-350 reactor.

Analysis of Heterogeneity Effects

The BFS assemblies have a laminated structure in the vertical direction which may have a great effect on the individual physical characteristics. Thus, the uranium-235 blocks are about 5 mm high; the uranium-238 blocks are 10 mm. The effects of the heterogeneous structure have been studied experimentally and theoretically on the 15-3 assembly. It should be noted that the heterogeneous structure of assembly 15-3 is a complex one: the

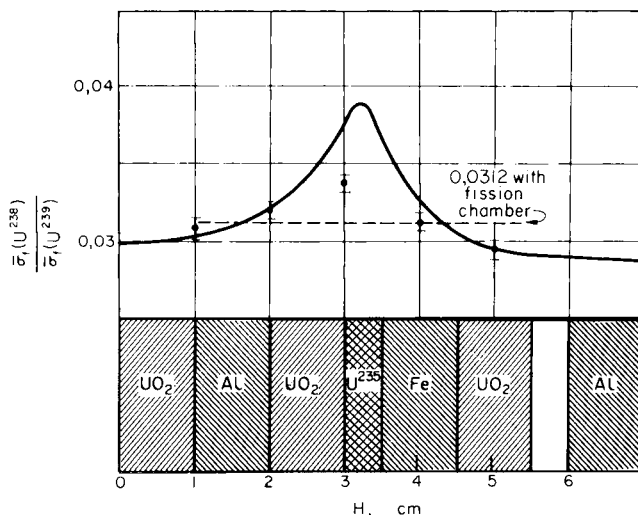


FIG. 8. Experimental and theoretical distribution of the average fission cross-section ratios of U-238 and U-235 ($\sigma_f^{-8}/\sigma_f^{-5}$) with respect to the height of a heterogeneous cell of the BFS-15 assembly (dots—experiment, continuous line—calculation, 18-group S_8 approximation was used). The broken line shows the measurement with fission chambers.

arrangement of the blocks is different in the high- and low-enrichment zones, in each of the zones the individual cells have different sequences of blocks of materials in the vertical direction. The results given below relate to the most typical elementary cells.

The experimental investigations consisted of detailed measurements of the fission reaction rates in the axial direction. The measurements were made by activation methods using scintillation gamma spectrometry.⁽⁶⁾

During the experiments, uranium foils 6 mm in diameter and 0.2 mm and 0.07 mm thick made of natural and 90% enriched uranium were placed between the blocks in the fuel channel. In certain experiments, in order to study the pattern of the reaction inside the blocks, the uranium blocks were made up of separate disks about 1.5 mm thick, between which the foils used for the measurements were inserted.

These experiments revealed the effect of heterogeneous arrangement of materials in the reactor core on the neutron flux distribution. The heterogeneity has the greatest effect on the distribution of the U-238 fission rate (Fig. 8). The weakest effect is on the distribution of the U-235 fission reactions where the values for the effects obtained (about 2%) scarcely exceeded the error of the measurements. The measurements inside a block of U-238 showed that the rates of the fission and capture reactions depend on the material of the adjacent blocks (Fig. 9). Substantial alteration of the reaction rates takes place in the relatively thin boundary layer.

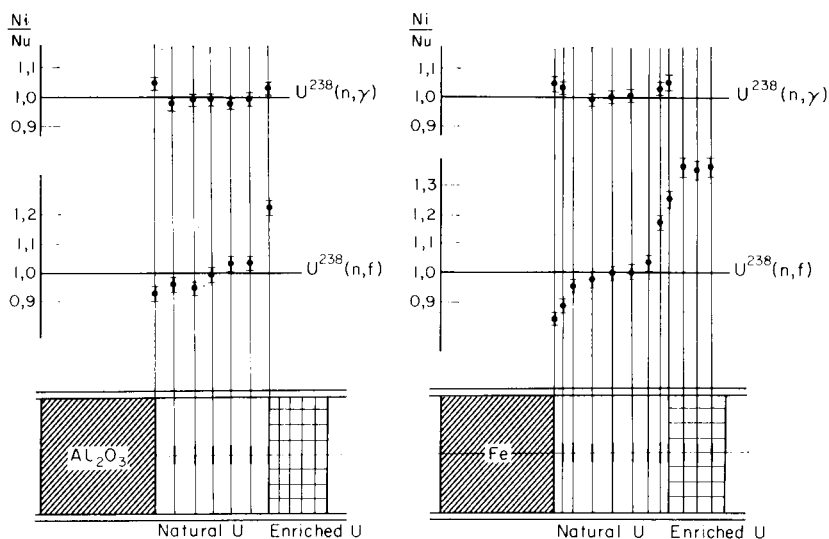


FIG. 9. Distribution of fission (n, f) and captures (n, γ) of U-238 through block of natural enriched uranium.

The heterogeneous structure of the assemblies is revealed in measurements of the axial fission distribution not only by foils but also by fission chambers with U-238, Np-237, Th-232. The chamber with U-235 reveals practically no irregularities in the fission distribution. For the 15-3 assembly in the low-enrichment zone where 36% enriched uranium is used, the irregularities of U-238 fission (for the chambers) lie within the errors. In the high enrichment zone the amplitude of the peaks amounts to 10% of the measured quantity.

It is clear that the space resolution of the above two methods of determining the distribution of U-238 fissions is different. The measurements with the chambers give a result which is averaged over the height of the chamber, while the measurements with the foils are related practically to a point.

During the measurements the two methods were compared at different

points of the core (using the ratio of the fission cross-sections of U-238 and U-235):

Foils	0.0338	0.0252
Chambers	0.0312	0.0242

These results show that when measuring the fission rate of threshold isotopes using chambers there may be some uncertainty due to the heterogeneous structure, making the interpretation of the experimental results difficult.

A theoretical analysis of the heterogeneous effects was made on the basis of the Sn method.

Allowing for the heterogeneous composition of the assembly leads to some softening of the theoretical spectrum. This can be seen from the following comparison of the neutron capture cross-section of boron-10:

Homogeneous calculation with initial cross-sections	$\sigma_c^{B^{10}}$	= 1.75
Homogeneous calculation with cross-sections averaged over the heterogeneous cell	$\sigma_c^{B^{10}}$	= 1.85

Calculations made with the group cross-sections averaged with respect to the flux in the cell have made it possible to assess the effect of heterogeneity in the reactivity.

It should be emphasized that as a result of the diversity of the cells such an assessment is fairly approximate. For the maximum assessment it was assumed that the whole core of the assemblies BFS-15 and BFS-16 has the structure of some average cell in the high-enrichment zone, where the heterogeneity is greater than in the low-enrichment zone.

In this case the correction to the theoretical values of K_{eff} is about -0.1% .

During the course of this study, attention was directed towards the part played by resonance on heterogeneity effects, due in particular to the block arrangement of U-238 oxide. With the aid of calculations it was found that the influence of this effect on the change in spectrum and reactivity can be ignored.

Prompt Neutron Lifetime

The prompt neutron lifetime was measured on the assemblies BFS-15-3 and BFS-16-1 by the Rossi- α method. The detectors used were two proportional counters with enriched BF_3 . The distribution of the two delayed coincidences was measured with a 10-channel time analyser with different channel widths, which made it possible to subtract the background of random coincidences more accurately. The degree of subcriticality was determined from the position of a control element calibrated in terms of ρ/β during start-up. The effective proportion of delayed neutrons determined by calculation was 0.007 in both cases.

To allow for the effect of the geometry of the detector position, measurements were made of the multiplication constant α and subcriticality for:

- (a) detectors located in the central region of the core,
- (b) detectors located in the outer layer of the reflector.

In addition to this, for both detector positions α was measured with a neutron source in the centre of the reactor. In all cases the values of α coincided within the limits of the error of measurement (about 5%).

The theoretical lifetime was determined from the formula:

$$= \frac{\int dv \left(\sum_i \frac{\phi_i \phi_i^*}{v_i} \right)}{\text{FNI}}$$

where ϕ_i , ϕ_i^* are the group fluxes and importance, v_i is the group mean velocity, FNI means fission neutron importance.

The experimental value of the lifetime was much greater than the theoretical one. These values are given in Table 9.

TABLE 9. PROMPT NEUTRON LIFETIME $\times 10^7$ sec

	BFS-15-3	BFS-16-1
Experiment	6 ± 0.3	4.93 ± 0.25
Calculation	3.4	3.1

The reasons for such great differences have not been definitely explained.

2. Calculations of the Physical Characteristics of ZPR-III Critical Assemblies

To check the system of constants used for the calculation of fast reactors, some experimental data obtained on ZPR-III critical assemblies were used. The most detailed and systematic analysis of these experiments is given in the work by Davey.⁽⁷⁾ For eleven of these assemblies, calculations were carried out using 26-group BNAB constants⁽²⁾ with the number of groups reduced to eighteen (see above). The calculations were made in diffusion approximation.

In ref. (8), detailed data are given on the cross-sections used for the calculation of critical assemblies and more detailed results of calculations than in the present paper.

Table 10 gives the composition and dimensions, calculated from the critical masses, of the sphericalized cores for the eleven critical assemblies.

To assess the accuracy of our diffusion calculations and check the initial data, three critical assemblies (Nos 11, 12, 24) were calculated with the YOM constants used in ref. (7). A comparison was made of the multiplication factors, fission cross-section ratios in the centre of the core, the reactivities introduced by specimens in the centre, the average cross-sections of boron-10 in the centre, the prompt neutron lifetime. In all the quantities, apart from the lifetime, good agreement between the results was obtained. The differences do not exceed 1% and in the majority of cases it is much lower, although in ref. (7) it is the S_4 approximation and not the diffusion approximation which is used. This means that the differences between the experimental and theoretical data obtained, and the difference in the results, arise through the inaccuracies in the system of constants and the difference between the

TABLE 11. FISSION CROSS-SECTION RATIOS IN THE CENTRE OF THE CORE

	σ_f^5/σ_f^0	σ_f^3/σ_f^0	σ_f^8/σ_f^0	σ_f^4/σ_f^0	σ_f^6/σ_f^0	σ_f^{40}/σ_f^0
$(E/C)_{av}$	0.988	0.990	1.009	1.044	1.028	0.985
$(E/C)_{max}$	1.012	1.015	1.065	1.138	1.27	1.101
	(23)*	(29)	(29)	(29)	(29)	(29)
$(E/C)_{min}$	0.958	0.948	0.97	0.954	0.637	0.855
	(31)	(24)	(11)	(25)	(33)	(31)

* In parentheses—number of critical assembly.

systems of constants, but not from error in the method of calculation. The exception is the prompt neutron lifetime, calculated in ref. (7) by the flux decay constant α , but which in this instance was calculated with the fluxes, importance, and mean group velocities (see above).

The main results of the calculations reduce briefly to the following.

The multiplication factors are given in Table 10. The values of k_{eff} which were calculated systematically exceed unity and the values obtained by Davey. For the eleven critical assemblies $k_{BNAB} = 1.015$ and $k_{YOM} = 1.002$.

In comparing the fission cross-section ratios in the centre of the assemblies, the refinements of the experimental values of these ratios made in ref. (9) were taken into account.

Table 11 compares the experimental and theoretical fission cross-section ratios. The average ratios of the experimental and theoretical values for some critical assemblies are given (E/C) , and also the deviations from the average.

The values obtained $(E/C)_{av}$ are fairly satisfactory, although in the case

of threshold detectors the deviation from the average is large, especially for U-236 and Pu-240.

The prompt neutron lifetime values obtained are closer to the experimental data than Davey's values. The mean value of the ratio of the experimental l to the theoretical was 1.18 (Davey has 1.31), the minimum was 1.11 (for No. 31), and the maximum 1.25 (for Nos. 12 and 34). As already mentioned, if the formula adopted here for l and the YOM constants are used, the experimental and theoretical results agree better, but even in this case the YOM constants give lower values for l than the BNAB constants, i.e. they predict a harder spectrum.

TABLE 12. RATIOS OF EXPERIMENTAL TO THEORETICAL CENTRAL REACTIVITY COEFFICIENTS, AVERAGED OVER A GROUP OF ASSEMBLIES

	Number of averaged values	Average E/C	Maximum E/C	Minimum E/C
U-235	11	0.915	0.947 (29)*	0.891 (11)
U-233	11	1.011	1.098 (29)	0.958 (24)
U-238	10	0.832	0.986 (30)	0.539 (23)
Th-232	6	0.756	0.807 (24)	0.696 (33)
Nb	7	0.827	0.909 (12)	0.74 (33)
Mo	8	0.866	0.96 (33)	0.73 (25)
Ta	8	0.886	0.968 (24)	0.772 (23)
B-10	9	1.066	1.087 (33)	0.998 (23)

A comparison has also been made of the central reactivity coefficients for different materials. Some of the results obtained are given in Table 12. Some of these results show a substantial difference between calculation and experiment. The reactivity introduced by U-235 is systematically larger in the calculation than in the experiment, the amount of difference being approximately the same as in the BFS-15-3 and BFS-16-1 assemblies.

The prediction of the reactivity introduced by U-238 and thorium is poor. In the calculations of the reactivity coefficients of niobium, molybdenum, and tantalum no allowance was made for the effect of resonance shielding of cross-sections in the specimens. This led to a certain exaggeration of the theoretical reactivity coefficients for these materials. However, as a preliminary assessment shows, allowance for shielding only partially reduces the differences. Reactivity introduced by boron is predicted much better by these calculations than by those with the YOM system of constants.

The effectiveness of the scatterers Na, O, C, Al, in these calculations is closer to experiment than in ref. (7), but the differences are still large.

TABLE 13 (continued)

	10	11	12	13	14	15	16	17	18
core	0.0 ³ 956	0.0 ³ 154	0.0 ⁵ 945	0.0 ⁶ 172	+0.0 ⁹ 993	-0.0 ¹ 943	-0.0 ¹ 2173	-0.0 ¹ 3101	-0.0 ¹ 6145
blanket	0.0 ² 481	0.0 ⁵ 503	0.0 ⁷ 846	-0.0 ⁸ 595	-0.0 ⁹ 553	-0.0 ¹ 0172	-0.0 ¹ 2109	-0.0 ¹ 3128	-0.0 ¹ 6203
c	0.516	0.0904	0.0 ² 604	0.0 ³ 122	0.0 ⁵ 110	0.0 ⁸ 418	-0.0 ¹ 0628	-0.0 ¹ 1586	0.0 ¹ 5160
b	0.0117	0.0 ² 180	0.0 ³ 114	0.0 ⁵ 307	0.0 ⁷ 779	0.0 ⁸ 132	0.0 ¹ 1439	0.0 ¹ 1211	0.0 ¹ 3197
c	0.0 ² 434	0.0 ² 110	0.0 ³ 100	0.0 ⁵ 290	0.0 ⁷ 548	0.0 ⁹ 659	0.0 ¹ 1669	0.0 ¹ 3952	0.0 ¹ 6804
b	0.0298	0.0 ² 521	0.0 ³ 393	0.0 ⁴ 119	0.0 ⁶ 298	0.0 ⁸ 622	0.0 ¹ 0653	0.0 ¹ 252	0.0 ¹ 3273
-	10.7	1.41	0.0710	0.0 ² 133	0.0 ⁴ 158	0.0 ⁶ 122	0.0 ⁹ 901	0.0 ¹ 391	0.0 ¹ 4287
c	0.682	0.119	0.0 ² 646	0.0 ⁴ 906	0.0 ⁶ 508	-0.0 ⁸ 868	-0.0 ⁸ 138	0.0 ¹ 3668	0
b	0.0106	0.0 ² 158	0.0 ³ 106	0.0 ⁵ 328	0.0 ⁷ 666	0.0 ⁹ 426	0.0 ¹ 0561	0.0 ¹ 183	0
1 c	0.0433								
1 b	0.0 ³ 277								
2 c	0.226								
2 b	0.0 ² 133								
3 c	0.486								
3 b	0.0 ² 278								
4 c	0.912								
4 b	0.0 ² 555								
5 c	1.81								
5 b	0.0154								
6 c	4.00								
6 b	0.0478								
7 c	4.04								
7 b	0.0557								
8 c	2.70								
8 b	0.0430								
9 c	1.46								
9 b	0.0275								

TABLE 14 (continued)

[illegible]

It is possible that for these specimens it is important to take into account both the shielding of the cross-sections and the change in the spectrum near the scattering specimen.

The possibilities of analysis of experiments on critical assemblies are greatly extended when using the recently developed perturbation theory for linear fractional functionals of the flux and bilinear fractional functionals of the flux and importance.^(10, 11) From the formulae of the perturbation theory it is possible to find variations of all the parameters measured in the critical experiment caused by some changes in the system of constants or composition of the assembly. For this, each assembly must be provided with a set of coefficients for perturbation theory formulae, linking the variations of all the parameters measured on it with the perturbations in the system of constants.

In multi-group diffusion approximation the formulae of perturbation theory take the following form:^(10, 11)

for variation of the fission cross-section ratio in the centre of the assembly

$$\sigma_f = \sum_j \sigma_f^j F_j(0) / \sum_j F_j(0),$$

$$\frac{\Delta(\sigma_f^p / \sigma_f^q)}{(\sigma_f^p / \sigma_f^q)} = \sum_j \left(\frac{\Delta\sigma_{fp}^j}{\sigma_f^p} - \frac{\Delta\sigma_{fq}^j}{\sigma_f^q} \right) F_j(0) + \sum_N \sum_\alpha \rho \alpha N \hat{L}_N [\Delta\alpha_\alpha, (F\psi_{pq}^+)],$$

where

$$\hat{L}_N [\Delta\sigma_\alpha, (F\psi_{pq}^+)] = \sum_j \{ 3\Delta\sigma_{ir, \alpha}^j (i_j i_{pq, j})_N^{\rightarrow \rightarrow +} - \Delta\sigma_{cfd, \alpha}^j (F_j \psi_{pq}^+)_N + \\ + \sum_{K=1}^{j-1} \Delta\sigma_\alpha^{kj} (F_K \psi_{pq, j}^+)_N + \Delta(v_j \sigma_f)_\alpha^j (F_j Q_{pq}^+)_N \};$$

for variation of the prompt neutron lifetime

$$\frac{\Delta l}{l} = \sum_N \sum_\alpha \rho \alpha N \left\{ \hat{L}_N [\Delta\sigma_\alpha, (F\psi_l^+ + \psi_l F^+)] + \sum_j \Delta(v_j \sigma_f)_\alpha^j \frac{(F_j Q^+)_N}{\text{FNI}} \right\};$$

for variation of the central reactivity coefficient which equals

$$R = \frac{1}{\text{FNI}} \hat{L}_0(\sigma_{0\delta p}, FF^+),$$

$$\frac{\Delta R}{R} = \frac{\hat{L}_0(\Delta\sigma_{0\delta p}, FF^+)}{\hat{L}_0(\sigma_{0\delta p}, FF^+)} + \\ + \sum_N \sum_\alpha \rho \alpha N \left\{ \hat{L}_N [\Delta\sigma_\alpha, (F\psi_R^+ + \psi_R F^+)] - \sum_j \Delta(v_j \sigma_f)_\alpha^j \frac{(F_j Q^+)_N}{\text{FNI}} \right\}.$$

In these formulae j and K are the number of the energy group, N is an index for the zone of the assembly, α is the isotope index, ρ is the concentration, FNI is the fission neutron importance.

The method of determining the functions ψ and ψ^+ is in refs. (10, 11). The coefficients for these formulae have been calculated for various ZPR-III assemblies. As an illustration, Tables 13–15 give these coefficients for assembly No. 11. For brevity the coefficients for $3\Delta\sigma_{tr}$, $-\Delta\sigma_{cfd}$, $\Delta\sigma^{kj}$, and $\Delta(v_f\sigma_f)$ are indicated by a , b , c , and d respectively. When using these coefficients the variations of the cross-sections must be substituted in the formulae in barns.

TABLE 16. COMPARISON OF DIRECT CALCULATIONS WITH PERTURBATION THEORY CALCULATION

Perturbation		$\Delta\sigma_{cfd}^5$ core	$\Delta\sigma_{tr}^{Fe}$ core and blanket	$\Delta\sigma_{kj}^8$ blanket	$\Delta(v_f\sigma_f)^5$ core
ΔK_{eff}	1	0.0058	-0.036	-0.0152	-0.0680
	2	0.00547	-0.03555	-0.0156	-0.0680
$\frac{\Delta(\sigma_f^5/\sigma_f^9)}{(\sigma_f^5/\sigma_f^9)}$	1	-0.03117	-0.03308	-0.00128	0.03253
	2	-0.03111	-0.03298	-0.00131	0.03235
$\Delta l/l$	1	-0.00752	-0.00110	-0.0259	0.137
	2	-0.00726	-0.00106	-0.0259	0.124
$\Delta R/R$	1	-0.00556	-0.00388	0.0714	0.106
(B-10)	2	-0.00543	-0.00374	0.0704	0.0974

1—Direct calculation.

2—Perturbation theory calculation.

In Table 16 the results of the calculations of Δk_{eff} , $\Delta l/l$ and $\Delta R/R$ for B¹⁰ by perturbation theory are compared with the results of direct calculations of the variations of these parameters with certain perturbations in the cross-sections for assembly No. 11.

The perturbations taken for Table 16 are random and arbitrary and are only used to show that the perturbation theory methods developed and brought to practical use make it possible to assess accurately the effects of the influence of the constants on the very different characteristics of the critical assemblies.

3. Investigation of Neutron Propagation in Uranium Dioxide and the Doppler Effect in Uranium-238

The purpose of these investigations was to check the U-238 cross-sections used at present in the resonant energy range. In this section the results of three groups of work are given.

The first relates to the study of the neutron spectrum from the interior of a solid prism of uranium dioxide. This work was carried out on the pulsed reactor IBR.

The second group of work is associated with the study of the space-energy neutron distribution in the thick oxide blanket of the reactor BR-1.

Finally, the third group of work consisted in the measurement of the influence of the Doppler effect on the amount of activation of uranium samples placed in the oxide blanket of the BR-1 reactor.

In all cases comparisons were made with the calculations and the possible reasons for differences were considered.

Investigations of the Neutron Spectrum in Uranium Dioxide by the Time-of-flight Method in the IBR Reactor

The system of measuring the neutron spectra from different media by the time-of-flight method using the pulsed fast reactor IBR as the source has been described in detail in ref. (12). The special feature of the experiment described consisted in the fact that an attempt was made to measure the spectrum of the neutrons existing inside the medium and not the escape spectrum through a plane surface, as was the case in the previous experiments.^(12, 13) For this purpose the uranium dioxide prism under investigation was made up of two parts, in the first of which, in the main, the formation of the neutron spectrum under investigation took place, while the second collimation part of the prism, of the same composition, had through holes for extraction of the neutrons in the direction of the neutron guide (Fig. 10). The porosity of the collimation part of the prism was 20%. The thickness of the part of the prism forming the spectrum was 35 cm; the thickness of the collimation part was 45 cm; the transverse dimensions of the prism were 80 × 80 cm. The resolution of the time-of-flight spectrometer was 5 μ sec/m. (Fig. 11).

Asymptotic equilibrium of the neutron spectrum in uranium oxide has not yet been established on the thicknesses investigated: the spectrum of the neutrons leaving the prism is much harder than asymptotic. However, the extent of this difference in the energy range about 1 keV is less than is found from the calculations. The difference observed between the theoretical and experimental spectra cannot be explained by the direct influence of the anisotropic porosity of the collimation part. Calculation shows that if the collimation part of the prism is completely removed, the neutron spectrum becomes harder, whereas the experimental spectrum is softer than the theoretical. The main differences between the theoretical and experimental spectra are observed in the energy range below about 200 eV, where the uranium cross-sections are known with some accuracy. Therefore the inaccuracy of the group constants used might be caused only by the error in averaging these cross-sections over the resonance characteristics. In this

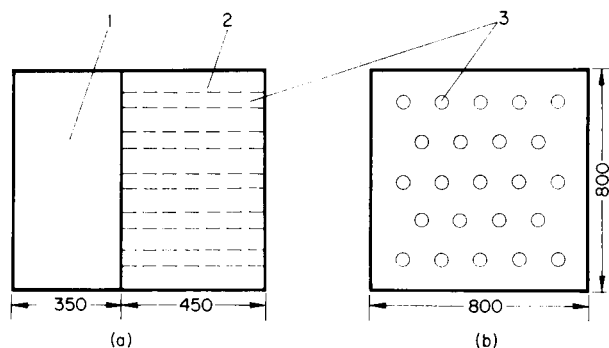


FIG. 10. Diagram of prism of uranium oxide.

- A — view from the side.
- B — view on collimator side.
- 1 — prism forming the spectrum.
- 2 — collimator.
- 3 — exit channels of collimator.

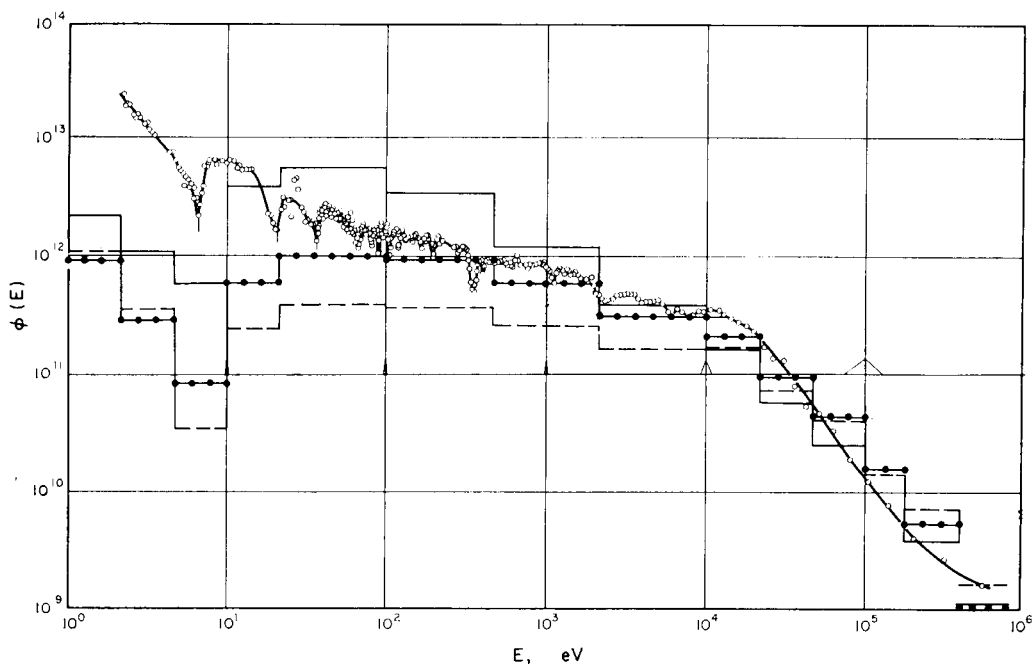


FIG. 11. Neutron spectrum in uranium oxide. Circles—experiment. Continuous histogram—theoretical asymptotic neutron spectrum in uranium oxide. Broken histogram—spectrum of leakage neutrons from prism 35 cm thick (no collimator). Histogram with black dots—total neutron spectrum leaving via the collimator.

averaging it was assumed that the detail structure of the total neutron flux is determined by the factor $1/\Sigma_{\text{tot}}(E)$, the structure of the neutron current spectrum by the factor $1/\Sigma_{\text{tot}}^2(E)$, the structure of the higher harmonic spectrum, because of the transport approximation used in the calculations, was assumed to be the same as the first harmonic. As was shown in ref. (14), and will be mentioned below, the actual situation is much more complex.

The evaluations, made in accordance with the following section, of the effect of the neutron flux gradient on the mean group constants show that the corrections produced by this effect are small. However, the clarification of the role of these corrections in the concrete case under consideration requires a special investigation, which at present is not yet complete.

Another purely experimental reason for the differences in question in the form of the spectra may be an impurity in the neutron beam under investigation, coming not from the orifices in the collimation part of the prism but from its surface. According to assessments this impurity amounts to about 10% and has a spectrum closer to the asymptotic than the spectrum inside the prism. By allowing for this effect it is possible to reduce the differences between the theoretical and experimental spectra, although it is not possible to eliminate them completely.

It should be noted that in the energy range below 200 eV where differences are observed, there is only a comparatively small proportion of the neutron spectrum (about 3%). At higher energies the agreement between the theoretical and experimental data is satisfactory.

Measurement of the Space-energy Neutron Distribution in the Thick Oxide Blanket of the BR-1 Reactor

The experiment on the BR-1 reactor with the oxide blanket is similar in arrangement and method to the experiments already described on propagation of neutrons in uranium carbide^(15, 16) (Fig. 12).

Along the radius of the blanket in the direction of its greatest thickness we measured the distribution of reaction rates with different cross-section energy dependence (Fig. 13).

As can be seen from the diagram, there are considerable differences between the experimental and theoretical results in the oxide blanket. The theoretical distributions are much steeper. These differences were somewhat unexpected because, for a uranium carbide blanket comparatively close in its nuclear physical characteristics to dioxide, the theoretical and experimental distributions agreed fairly well with one another.⁽¹⁶⁾ It may be thought that the differences observed are caused by the low energy part of the spectrum, the role of which in the uranium dioxide blanket is much greater than in the spectra established in uranium carbide. This is also indicated by a comparison of the theoretical and experimental values of the

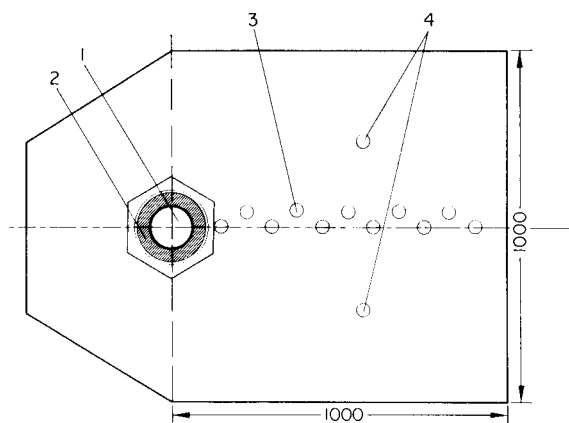


FIG. 12. Diagram of BR-1 reactor with oxide blanket.

- 1 — core.
- 2 — control of elements.
- 3 — channels for measuring reaction distribution.
- 4 — channels for measuring Doppler effect.

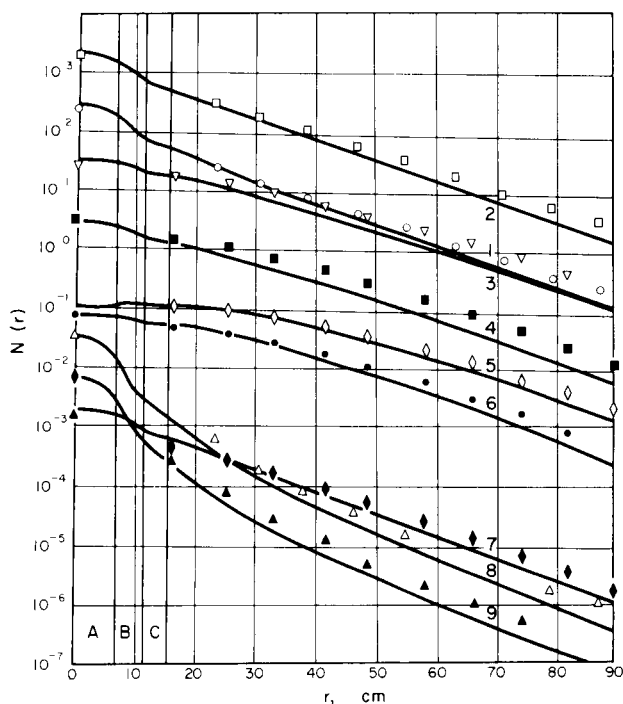


FIG. 13. Distribution rates of different reactions—oxide blanket of BR-1 reactor.

- A — core.
- B — control elements.
- V — transition zone.
- C — oxide blanket.

Dots—experiment. Curves—results of calculation in transport P_3 approximation.
Geometry—one-dimensional (sphere).

ratios of the cross-sections of various reactions to the fission cross-sections of plutonium-239 (see Table 17).

TABLE 17. RATIO OF THE REACTION CROSS-SECTIONS IN THE SPECTRUM ESTABLISHED IN THE OXIDE BLANKET OF THE BR-1 REACTOR AT A DISTANCE OF 68 CM FROM ITS CENTRE

Reaction	σ reaction/ σ_f (Pu-239)	
	Experiment	Calculation
U-235 (n, f)	1.59 ± 0.14	1.21
Au-197 (n, γ)	1.4 ± 0.1	1.07
Na-23 (n, γ)	0.0032 ± 0.0003	0.00225

Measurement of the Doppler Effect in Uranium Dioxide

In the oxide blanket of the BR-1 reactor in two symmetrically placed channels at a distance of 68.3 cm from the centre of the reactor, a measurement was made of the influence of the Doppler effect on the activation of uranium specimens. Measurements specially made showed that the neutron fluxes in places where irradiation of specimens was being carried out differ from one another by no more than 0.4%. At the same time two specimens of U_3O_8 were irradiated in thin-walled stainless steel containers surrounded by a nichrome wire heater and fitted with a thermocouple. One specimen was irradiated at 300°K, the other at 800°K.

The size of the specimen was chosen so that the mean free path of the neutrons captured in the uranium

$$l = \frac{\langle \Sigma_c(U^{238}) / \Sigma_{tot} \rangle}{\langle \Sigma_c(U^{238}) \rangle}$$

was much larger than the mean thickness of the specimen

$$d = \frac{4V}{S}.$$

In the present case the ratio d/l was less than 0.2 in all energy ranges, making no substantial contribution to the activation of the specimen. This fact made it possible to consider the specimen thin.

The activities of weighed samples of uranium oxide from the irradiated specimens were measured from the gamma line of uranium-239 with energy of 74 keV. As a result of a series of several measurements the ratio of the specific activities of the hot and cold specimen was determined and was found to be 1.09 ± 0.01 .

To determine the theoretical magnitude of this effect, group capture cross-sections of the hot specimen irradiated in a cold medium were specially compiled (see section 4). When using these constants the above ratio was 1.15. This far exceeds the experimental value. The reasons for this difference are being analysed.

It must be noted that the Doppler effect in a thin specimen placed in a cold medium is very different from the Doppler effect for a large specimen (or for a specimen placed in a heated medium). For strong resonances (low energies) the Doppler effect is greater than homogeneous for the small specimen, and for weak resonances (high energies) it is smaller. Therefore, depending on the neutron spectrum, the total effect for the small specimen may be either more or less than the Doppler effect in a homogeneous medium.

In our case calculation gives a Doppler effect of about 1.12 for the large specimen compared with 1.15 for the small specimen.

4. Some Problems of Calculation Theory and Methods

Effective Method of Calculating Nuclear Reactors

In theoretical investigations of fast reactors for the purpose of optimization, wide use is made of a combination of computer programmes, the algorithms for which are based on the use of effective synthetic methods of calculating neutron fluxes.

Let us write the equation for transfer of neutrons in a quasi-critical reactor in operator form:

$$\hat{L}\phi(\xi, \eta) = \frac{1}{K_{\text{eff}}}\hat{Q}\phi(\xi, \eta) \quad (1)$$

where $\hat{L}$ and $\hat{Q}$ are operators corresponding to that type of equation, the solution of which has to be found; $\phi(\xi, \eta)$ is the neutron flux defined in the phase space of the variables ξ, η where ξ is a space co-ordinate (x, y , or r), and η can represent an energy, angular, or space variable. The problem is also easily generalized to a greater number of independent variables.

In the effective method of resolution, equation (1) is converted to a system of equations with respect to functions depending only on one of the arguments ξ and η . For this purpose the phase space (ξ, η) by dividing the co-ordinate axes into a series of intervals $\Delta\xi$; and $\Delta\eta_K$ is divided into bands within which we search for the distributions

$$I_K(\xi) = \int_{\Delta\eta_K} \phi(\xi, \eta) d\eta; \quad \psi_i(\eta) = \int_{\Delta\xi_i} \phi(\xi, \eta) d\xi. \quad (2)$$

TABLE 18

Designation of programme Number of energy groups Geometry Which variables are conditionally separated Number of zones with separate variables to total number of zones Calculation time with accuracy of ± 0.001 with total number of zones and 1000 elements	18-1-RZ-01K*	18-RZ-01C to 18 [†]	18-RZ-36 to 18	18-4-RZ-15 1+4 [‡]	26-X _μ -1200 to 26
	I* RZ $r-z-E$ $\frac{1}{4}$ ~ 10 sec	RZ $r-z$ 1 $\frac{1}{25}$ ~ 5 min	RZ $r-z$ 36 $\frac{36}{36}$ 10-20 min	RZ $r-z-E$ 15 $\frac{15}{15}$ 3-7 min	X $x-\mu$ 1200 $\frac{1200}{1200}$ ~ 4 min

* Programme carries out a complete calculation of a fast reactor,⁽²⁰⁾ With the aid of the importance, the problem reduces from 18-group to 1-group.

† Problem reduced from 18-group to a problem with 1 to 4 groups.

‡ Programme allows neutron importance to be calculated.

Equation (1), assuming division of the variables in each region $G_{i, K}$ of the phase space (ξ, η) , which is the area of intersection of the bands $\Delta\xi_i$ and $\Delta\eta_K$ can be reduced to the system

$$\hat{L}_\psi \Phi_K(\xi) = \frac{1}{K_{\text{eff}}} \hat{Q}_\psi \Phi_K(\xi); \quad \hat{L}_{\phi\psi_i}(\eta) = \frac{1}{K_{\text{eff}}} \hat{Q}_{\phi\psi_i}(\eta) \quad (3)$$

where the operators $\hat{L}_\psi$, $\hat{Q}_\psi$ and $\hat{L}_\phi$, $\hat{Q}_\phi$ contain functionals defined on the space of the functions $\psi(\eta)$ and $\phi(\xi)$, respectively. If $\hat{L}$ is a differential operator, then using the conditions of continuity of the neutron flux and current, in equation (3) it is possible to include the distribution functions in adjacent bands.

$$(\phi_{K-1}, \phi_{K+1} \text{ and } \psi_{i-1}, \psi_{i+1}).$$

The systems developed for joint resolution of equation (3) ensure fairly rapid convergence of the iteration count. The method described above is used for the calculation of two-dimensional reactors in multi-group approximation,⁽¹⁷⁾ in a programme of effective low-group calculation⁽¹⁸⁾ and in calculation of the space-energy distribution of neutrons.⁽¹⁹⁾

On the basis of the functions found $\phi(\xi)$ and $\psi(\eta)$ it is possible to synthesize the detailed neutron distribution in a class of functions which are discontinuous at the zone boundaries with separated variables, and in a class of continuous functions.⁽²¹⁾ The results of calculations of the various integral characteristics (k_{eff} , conversion ratios, and non-uniform heat production, etc.) using the effective method, show good agreement with the results of exact calculations even with a comparatively small number of zones with separate variables. There is a considerable saving in computer time. Table 18 gives an illustration of the characteristics of programmes for computers using the idea of the effective method.

Theory of Neutron Propagation in Media with Resonance and Average Cross-sections

The transport equation for the resonance-averaged neutron flux $\bar{F}(x, E, \mu)$ is written in ref. (14). In L-representation it takes the form

$$\bar{\Sigma}_l \bar{F}_l + \frac{1}{2l+1} \left\{ (l+1) \frac{\partial \bar{F}_{l+1}}{\partial x} + l \frac{\partial \bar{F}_{l-1}}{\partial x} \right\} = \int dE' \bar{F}_l(x, E') \bar{\Sigma}_{sl}(E' \rightarrow E) + q(x, E) \quad (1)$$

where

$$\bar{\Sigma}_l = \frac{\langle \Sigma F_l \rangle}{\langle F_l \rangle}, \quad \bar{\Sigma}_{sl}(E' \rightarrow E) = \frac{\langle F_l(E') \Sigma_{sl}(E' \rightarrow E) \rangle}{\langle F_l(E') \rangle};$$

here the sign $\langle \rangle$ means ordinary averaging with respect to energy and weight l . The spectrum of the angular moments $F_l(E)$ for averaging cross-sections can be found as an approximation, not taking into account the effect of the higher harmonics on the spectrum of low moments and the space variation of the spectrum. With respect to $F_0(E)$ this means approximation of the collision density constant.

In this case

$$F_l(E) \simeq \sum_{n=0}^l \frac{A_{l-n}}{\Sigma^{n+1}(E)},$$

$$A_l = \frac{\sum_{n=1}^l A_{l-n} \Sigma^{-n-1}(E) \bar{\Sigma}_{sl}(E)}{1 - \Sigma^{-1}(E) \Sigma_{sl}(E)}, \quad (2)$$

$$\Sigma_{sl}(E) = \int \Sigma_{sl}(E \rightarrow E') dE'. \quad (3)$$

Summation of equations (1) with respect to l with the weight $\frac{2l+1}{4R} \text{Pe}(\mu)$ leads to the transport equation for $\bar{F}(x, E, \mu)$:

$$\mu \frac{\partial \bar{F}}{\partial X} + \bar{F} \Sigma_{\min} = \int dE' d\vec{\Omega}' \bar{F}(x, E', \mu') \Sigma_s(E' \rightarrow E, \mu_0) -$$

$$- \int d\vec{\Omega}' \bar{F}(x, E, \mu') \Sigma'(E, \mu_0) + q(K, E, \mu) \quad (4)$$

where

$$\bar{\Sigma}_s(E' \rightarrow E, \mu_0) = \sum_{l=0}^{\infty} \frac{2l+1}{4\pi} \bar{\Sigma}_s(E' \rightarrow E) P_l(\mu_0)$$

$$\Sigma'(E, \mu_0) = \sum_{l=0}^{\infty} \frac{2l+1}{4\pi} [\bar{\Sigma}_l(E) - \Sigma_{\min}] P_l(\mu_0) \quad (5)$$

and $\Sigma_{\min}$ is the minimum value of the total cross-section in the averaging interval. Thus, in the resonance-averaged transport equation, together with $\Sigma_{\min}$, there appears a term recalling in form the scattering integral with differential cross-section $\Sigma'(E, \mu_0)$.

Apart from the assumptions indicated above, it was assumed that the resonances are narrow and that the Placzek oscillations of the spectrum after averaging do not play an important part.

The limits of applicability of these equations and their accuracy can be assessed by comparison with the exact resolution of the homogeneous transport equation for the exponential space function:

$$F(x, E, u) = \phi(E, \mu) e^{-kx}. \quad (6)$$

The neutron flux spectrum is defined by the equation

$$F(E, \mu) = \frac{1}{\Sigma(E) - k\mu} F(E', \mu') \cdot \Sigma_s(E' \rightarrow E, \mu_0) dE' d\vec{\Omega}'; \quad (7)$$

integration of (7) with respect to $\vec{\Omega}$ with $\text{Pe}(\mu)$ leads to the system

$$\Sigma(E) F_l(E) = \sum_{l=0}^{\infty} Q_{ll} \left(\frac{K}{\Sigma(E)} \right) \int F_{l'}(E') \Sigma_{sl'}(E' \rightarrow E) dE' \quad (8)$$

where

$$Q_{ll'} \left(\frac{K}{\Sigma(E)} \right) = \frac{2l+1}{4\pi} \int \frac{\text{Pe}(\mu) \cdot P_l(\mu)}{1 - K\mu/\Sigma} d\vec{\Omega}. \quad (9)$$

From (9) it is clear that if

$$|K| < \Sigma_{\min}, \quad (10)$$

then $Q_{ll'}$ can be expanded into a power series with respect to $\frac{k}{\Sigma(E)}$, so that the spectrum of moments F_l is

$$F_l(E) = \sum_{n=0}^{\infty} \frac{A_{nl}}{\Sigma^{n+1}(E)} \quad (11)$$

where the coefficient A_n does not now depend on k . Where $k \rightarrow 0$ the expressions (11) convert into (2) with an accuracy to a power of about k .

Thus the microspectrum of the neutron flux $\text{Fe}(E)$ can be represented by expansion in functions $\frac{1}{\Sigma^n(E)}$, if condition (10) is fulfilled. Then the average cross-sections can be expressed by integrals of the type

$$\int \frac{\Sigma_x(E)}{\Sigma^n(E)} dE.$$

In the general case we can present the solution in the form of superposition of functions (6) (Fourier integral). Close to concentrated sources or boundaries between different media the contribution to the Fourier integral is given by high values of k , and therefore the expansion (11) is inapplicable here. Physically it is clear that in these regions there is a transient spectrum depending both on the cross-sections of the given medium and on the adjacent medium. Therefore an exact description of the flux near boundaries and sources requires a special theory. In the asymptotic region, however, at distances $x \gtrsim \lambda$ the condition (10) is fulfilled and the spectrum $\text{Fe}(E)$ can be presented in the form (11). The slow space variation of the spectrum in this case can be taken into account by introducing coefficients A_n depending on x .

A particularly simple form is assumed by equation (8) if the neutron scattering is isotropic ($\Sigma_{sl} = 0$ where $l > 0$).

Then

$$\Sigma(E) \cdot F_0(E) = Q_0 \left(\frac{k}{\Sigma(E)} \right) \int F_0(E') \Sigma_{s0}(E' \rightarrow E) dE'$$

$$F_l(E) = \frac{Q(K/\Sigma)}{Q_0(K/\Sigma)} F_0(E) \quad (12)$$

where

$$Qe(x) = \frac{1}{2} \int_{-1}^1 \frac{P_l(\mu) d\mu}{(1-x)\mu};$$

in particular

$$Q_0 = \frac{\Sigma(E)}{2K} \ln \frac{1+K/\{\Sigma(E)\}}{1-K/\{\Sigma(E)\}}.$$

Formula (12) shows that diffusion leads to non-constant collision density

$$F_0(E) \cdot \Sigma(E) \sim Q_0 \left(\frac{k}{\Sigma(E)} \right)$$

and, where $k^2 > 0$, the collision density increases comparatively in regions with minimum cross-sections, which must lead to some reduction in the average resonance cross-sections, in particular the absorption cross-sections (where $k^2 < 0$, on the other hand, the spectrum is enriched by the resonance neutrons).

In this case the average cross-sections Σ_l appearing in equations (1) and (4) equal

$$\bar{\Sigma}_l = \frac{\langle Q_l(K/\Sigma) \rangle}{\langle Q_l \Sigma^{-1} \rangle} \quad \text{instead of} \quad \bar{\Sigma}_l = \frac{\langle \Sigma^{-l} \rangle}{\langle \Sigma^{-l-1} \rangle};$$

as a first approximation with isotropic scattering. Let us calculate these quantities assuming high Breit-Wigner resonances:

$$\Sigma(E) = \Sigma_p + \sum_i \frac{\Sigma_{r0}^i}{1 + \{E - E_i/(\Gamma_i/2)\}^2}.$$

In this case

$$\left\langle Q_l \left(\frac{K}{\Sigma} \right) \right\rangle = \frac{1}{2} \int_{-1}^1 P_l(\mu) d\mu \left\{ \frac{1}{1 - (K/\Sigma_p)\mu} - u \frac{\Sigma_r}{\Sigma_p} \frac{K}{\Sigma_p} \frac{1}{\{1 - (K/\Sigma_p)\mu^{3/2}\}} \right\}$$

$$\left\langle \Sigma^{-1} Q_l \left(\frac{K}{\Sigma} \right) \right\rangle = \frac{1}{2\Sigma_p} \int_{-1}^1 P_l(\mu) d\mu \left\{ \frac{1}{1 - (K/\Sigma_p)u} - \frac{\Sigma_r}{\Sigma_p} \frac{1}{(1 - (K/\Sigma_p)\mu)^{3/2}} \right\} \quad (13)$$

where

$$\begin{aligned}\tilde{\Sigma}_r &= \frac{\pi}{2} \frac{\Gamma \Sigma_{r0}}{D} \sqrt{\left(\frac{\Sigma_p}{\Sigma_{r0} + \Sigma_p} \right)} \\ \Sigma^{-l} &= \frac{1}{\Sigma_p^l} \left[1 - \frac{\tilde{\Sigma}_r}{\Sigma_p} A_l \right] \\ A_l &= \sum_{n=1}^l (-1)^{n-1} C_n^l \frac{(2n-3)!!}{(2n-2)!!},\end{aligned}$$

C_n^l are binomial coefficients.

With low values of

$$\frac{\tilde{\Sigma}_r}{\Sigma_p} \text{ and } \frac{K}{\Sigma_p},$$

expanding with respect to these quantities, we find in particular for Σ_0 and $\bar{\Sigma}_1$:

$$\begin{aligned}\bar{\Sigma}_0 &\simeq \Sigma_p + \tilde{\Sigma}_r \left[1 - \frac{5}{24} \left(\frac{K}{\Sigma_p} \right)^2 \right] \\ \bar{\Sigma}_1 &\simeq \Sigma_p + \frac{\tilde{\Sigma}_r}{2} \left[1 - \frac{9}{40} \left(\frac{K}{\Sigma_p} \right)^2 \right].\end{aligned}\quad (14)$$

The formulae (15) show that the reduction (where $K^2 > 0$) of the average resonance cross-section due to deviation of the collision density from the constant value as a result of diffusion is slight (in the problem with neutron leakage ($K^2 < 0$) we find a corresponding increase in the average resonance cross-section). It should be noted, however, that in those cases when interference minima play a substantial part in the cross-sections, the effect of diffusion on the average cross-section increases.

Above we saw that the parameters of the transport equation, averaged with respect to the resonances, can be expressed in terms of integrals of the form:

$$\int \frac{\Sigma_x(E)}{\Sigma^n(E)} dE. \quad (15)$$

Until now it was assumed that the resonances are narrow compared with the neutron energy loss in scattering by each of the isotopes of the mixture. Nonfulfilment of this condition, like diffusion, leads to a deviation of the collision density from the constant value and ultimately to dependence of the average cross-sections on the mass of the absorber and moderator nuclei. Approximate examination shows that the finite value of the resonance width can be taken into account by introducing into the average cross-sections,

instead of the true values of the neutron width Γ_n and potential scattering cross-section σ_p , lower values:

$$\begin{aligned}\sigma_p^* &= \sigma_p \left[1 - \frac{1}{\lambda} \arctan \lambda \right] \\ \Gamma_n^* &= \Gamma_n \left[1 - \frac{1}{\lambda} \arctan \lambda \right]\end{aligned}\quad (16)$$

where

$$\lambda = \frac{2\xi E_0}{\Delta E},$$

ξ is the mean logarithmic energy loss for the isotope in question, ΔE is the energy interval inside which $\Sigma_r > \Sigma_p$ (the practical resonance width). Where $\lambda \rightarrow \infty$ we return to the narrow resonances, where $\lambda \rightarrow 0$ we arrive at the approximation of the infinite mass of absorber or scatterer.

It is shown in a number of works^(22, 23) that overlapping of resonances exerts a great influence on the average resonance cross-sections, in particular their temperature dependence. The case of weak or strongly overlapped resonances, when the resonance cross-sections can be considered as small fluctuations and the resonance shielding factor is expressed in terms of cross-section dispersion, has been studied in detail (see refs. (23, 24)). In the low-energy region this view is not correct. To calculate the average cross-sections in the general case we use the smallness of the ratio Γ/D and take into account the resonance overlapping as a first non-vanishing approximation with respect to Γ/D . We shall try to express the average cross-sections in terms of integrals of isolated resonance functions. The average cross-section $\bar{\Sigma}_x$ is expressed in terms of integrals of the form

$$\frac{1}{\Delta E} \int_{\Delta E} \frac{\Sigma_x(E)}{\Sigma(E)} dE,$$

$E_x(E)$ is the sum of the resonance cross-sections, and summation can be replaced by averaging with respect to distributions of the resonance parameters:

$$\frac{1}{\Delta E} \int_{\Delta E} \frac{\Sigma_x}{\Sigma} dE - \frac{1}{D} \left\langle \int \frac{\Sigma_x^i}{\Sigma} dx \right\rangle$$

where Σ_x^i is the cross-section corresponding to the i th resonance;

$$\Sigma = \Sigma_p + \Sigma_r^i + \Sigma_r^1$$

where Σ_r^i is the resonance cross-section without taking the i th resonance into account, then

$$\frac{1}{\Sigma} = \frac{1}{\Sigma_p + \Sigma_r^i} \left[1 - \frac{\Sigma_r^1}{\Sigma_p + \Sigma_r^i + \Sigma_r^1} \right]$$

and

$$\left\langle \int \frac{\Sigma_x^i}{\Sigma} dE \right\rangle = \left[\left\langle \int \frac{\Sigma_x^i}{\Sigma_p + \Sigma_r^i} dE \right\rangle - \left\langle \int \frac{\Sigma_x^i \Sigma_r^1 dE}{(\Sigma_p + \Sigma_r^1)(\Sigma_p + \Sigma_r^i + \Sigma_r^1)} \right\rangle \right]. \quad (17)$$

The second term describes the mutual shielding of resonances. We can also convert this, writing Σ_r^1 in the form of a sum of resonances excluding the i th

$$\Sigma_r^1 = \sum_{j=1} \Sigma_r^j, \quad \langle \quad \rangle = \sum_{j=1} \int \frac{\Sigma_x^i \Sigma_r^j dE}{(\Sigma_p + \Sigma_r^i)(\Sigma_p + \Sigma_r^i + \Sigma_r^j)}.$$

Transforming

$$\frac{1}{\Sigma_p + \Sigma_r^i + \Sigma_r^j} = \frac{1}{\Sigma_p + \Sigma_r^i + \Sigma_r^j + \Sigma_r''} = \frac{1}{\Sigma_p + \Sigma_r^j + \Sigma_r^i} \left[1 - \frac{\Sigma_r''}{\Sigma} \right],$$

where Σ_r'' is the resonance cross-section without the i th and j th resonances, we can continue this procedure, corresponding to expansion in a power series of Γ/D . Considering $\Gamma/D \ll 1$ we shall restrict ourselves to an examination of the paired interaction of resonances, ignoring Σ_r'' .

We make the following transformation

$$\begin{aligned} \frac{1}{\Sigma_p + \Sigma_r^i + \Sigma_r^j} &= \frac{1}{\Sigma_p + \Sigma_r^j} \left[1 - \frac{\Sigma_r^i}{\Sigma} \right] = \dots \\ &= \frac{1}{\Sigma_p + \Sigma_r^j} \left[1 - \frac{1}{\Sigma_p + \Sigma_r^i} \left[1 - \frac{\Sigma_r^j}{\Sigma_p + \Sigma_r^j} \left[1 - \dots \right] \right] \right]. \end{aligned}$$

Thus the correlation term in (17) can be written in the form of a sum of integrals of the type

$$\int \frac{\Sigma_r^{in} \Sigma_r^{jm}}{(\Sigma_r^i + \Sigma_p)^n (\Sigma_r^j + \Sigma_p)^m} dE. \quad (18)$$

Here it is possible to distinguish two cases: the resonances j belonging to the same system of resonances as the i th resonance (the same l, y , we shall represent by j^1) and the resonances of other systems (l, y) and other isotopes (j^{11}).

In the first case the distance between the i th and j th resonances is about D and small distances are improbable. Therefore the integral (18) has an order of magnitude $(\Gamma/D)^{2K}$, where K is the minimum of the numbers (n, m) . We ignore the quantities about $(\Gamma/D)^{2K}$ where $K > 1$. In addition, the

integrant in (18) where $E_i - E_j \sim D \gg \Delta E_{\text{eff}}$ takes the form of a double-humped curve with maxima where $E = E_i$ and $E = E_j$. Therefore, considering that $\Sigma_r^j(E_j \pm D) \ll \Sigma_p$ we can present it approximately in the form:

$$\frac{\Sigma_r^{jm}(E_i - E_j)}{\Sigma_p^m} \cdot \int \frac{\Sigma_r^{in}}{(\Sigma_r^i + \Sigma_p)^n} dE + \frac{\Sigma_r^{in}(E_i - E_j)}{\Sigma_p^n} \int \frac{\Sigma_r^{jm}}{(\Sigma_p + \Sigma_r^j)^m} dE.$$

In the second case the distances between the i th and j th resonances are distributed randomly and we can replace summation with respect to j by integrations with respect to dE_j^{11}/D^{11} where D^{11} is the mean distance between the levels j^{11} .

Here

$$\begin{aligned} \sum_{j^{11}} \int \frac{\Sigma_r^{in} \Sigma_r^{jm} dE}{(\Sigma_p + \Sigma_r^i)^n (\Sigma_p + \Sigma_r^j)^m} &= \int \frac{dE_j^{11}}{D_j^{11}} \int dE \frac{\Sigma_r^{in} \Sigma_r^{jm}}{(\Sigma_p + \Sigma_r^i)^n (\Sigma_p + \Sigma_r^j)^m} \\ &= \frac{1}{D_j^{11}} \int dE \frac{\Sigma_r^{j^{11}m}}{(\Sigma_p + \Sigma_r^{j^{11}})^m} \cdot \int dE (\Sigma_p + \Sigma_r^i)^n, \end{aligned}$$

i.e. these quantities are expressed in terms of the products of integrals, each of which is taken for an isolated resonance.

Thus, taking into account only the quantities $\geq (\Gamma/D)^2$, we can calculate the cross-sections of interest to us with overlapping of levels in terms of the integrals of isolated resonance functions:

$$\begin{aligned} \left\langle \int \frac{\Sigma_x}{\Sigma} dE \right\rangle_{ly} &\simeq \left\langle \int \frac{\Sigma_x^i}{\Sigma_p + \Sigma_r^i} dE \right\rangle - \frac{1}{\Sigma_p} \left\{ \left\langle \int \Sigma_r(E) P(E) dE \right\rangle_{ly} \left[\left\langle \int \frac{\Sigma_x}{\Sigma_p + \Sigma_r} dE \right\rangle_{ly} - \right. \right. \\ &\quad \left. \left. - \left\langle \int \frac{\Sigma_x \Sigma_r}{(\Sigma_p + \Sigma_r)^2} dE \right\rangle_{ly} \right] + \left\langle \int \frac{\Sigma_r}{\Sigma_p + \Sigma_r} dE \right\rangle_{ly} \left\langle \int \Sigma_x(E) P(E) dE \right\rangle_{ly} \right\} - \\ &\quad - \sum_{(ly)^1} \frac{1}{D_{(ly)^1}} \sum_{n=0} \left\langle \int \frac{\Sigma_r^{n+1}}{(\Sigma_p + \Sigma_r)^{n+1}} dE \right\rangle_{(ly)^1} \left[\left\langle \int \frac{\Sigma_x \Sigma_r^{n+1}}{(\Sigma_p + \Sigma_r)^{n+1}} dE \right\rangle_{ly} - \right. \\ &\quad \left. - \left\langle \int \frac{\Sigma_x \Sigma_r^{n+1}}{(\Sigma_p + \Sigma_r)^{n+2}} dE \right\rangle_{ly} \right]. \end{aligned} \quad (19)$$

Here $\langle \rangle_{ly}$ is averaging with respect to the parameters of the system (l, y) , $\langle \rangle_{(ly)^1}$ is averaging with respect to the parameters of all the other systems of levels apart from (l, y) , $P(E)dE$ is the probability of finding the level of the system (l, y) in the interval dE at a distance E from the isolated level of this system.

A similar examination is also carried out for integrals of the type

$$\int \frac{\Sigma_x}{\Sigma^n} dE.$$

Some of the integrals written down here were tabulated for the case of a

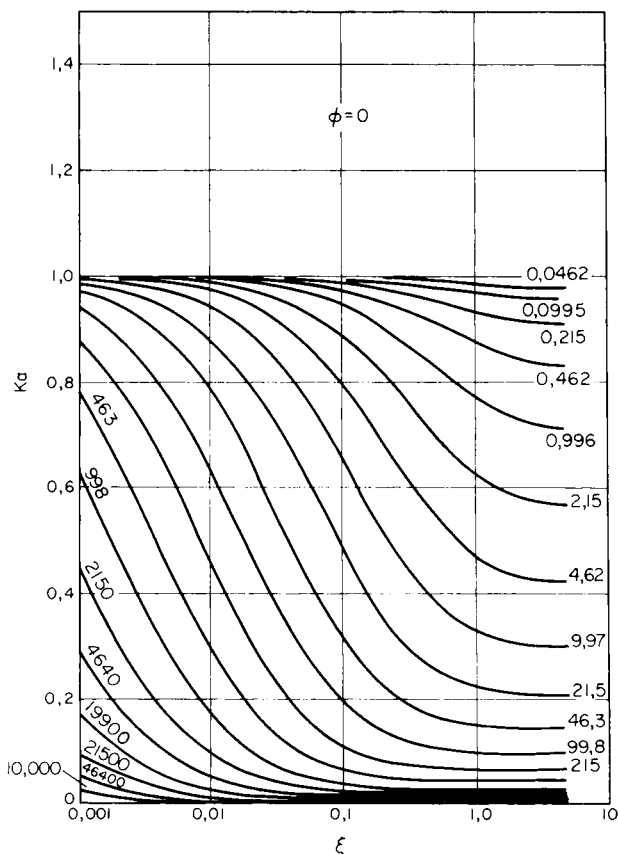


FIG. 14. Function $k_a(\xi)$ versus parameter $a = 0_r/0_p$ in the absence of interference of resonance and potential scattering ($\phi = 0$).

narrow isolated Breit-Wigner resonance. Using the notation of ref. (14), we write:

$$\left\langle \frac{1}{(\sigma_p + \tilde{\sigma})^n} \right\rangle = \frac{1}{(\sigma_p + \tilde{\sigma})^n} \sum_{m=0}^n (-1)^m C_n^m I_{rm}; \quad \left\langle \frac{\sigma_x}{\sigma + \tilde{\sigma}} \right\rangle = I_x;$$

where

$$\begin{aligned} I_{r,m} &= \frac{1}{\Delta U} \cdot \frac{\pi \Gamma \alpha}{2 E_0} \frac{1}{\pi} \int_{-\infty}^{\infty} \left[\frac{\psi(x, \xi) \cos 2\phi + \chi(x, \xi) \sin 2\phi}{1 + \alpha[\psi(x, \xi) \cos 2\phi + \chi(x, \xi) \sin 2\phi]} \right]^m dx \\ &= \frac{1}{\Delta U} \cdot \frac{\pi \Gamma \alpha}{2 E_0} K \\ I_x &= \frac{1}{\Delta U} \cdot \frac{\pi \Gamma \alpha}{2 E_0} \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{\psi(x, \xi) dx}{1 + \alpha[\psi(x, \xi) \cos 2\phi + \chi(x, \xi) \sin 2\phi]} \\ &= \frac{1}{\Delta U} \cdot \frac{\pi \Gamma \alpha}{2 E_0} \text{npu } x = \frac{E - E_0}{\Gamma/2}. \end{aligned}$$

The functions K_1 , K_2 , K_3 , K_4 and K_a were tabulated as functions of the parameters $\alpha = \frac{\sigma_{ro}}{\sigma_p + \sigma}$, $\xi = \Gamma/D$ and $\phi = KR$ (ref. 25) (Figs. 14 and 15).

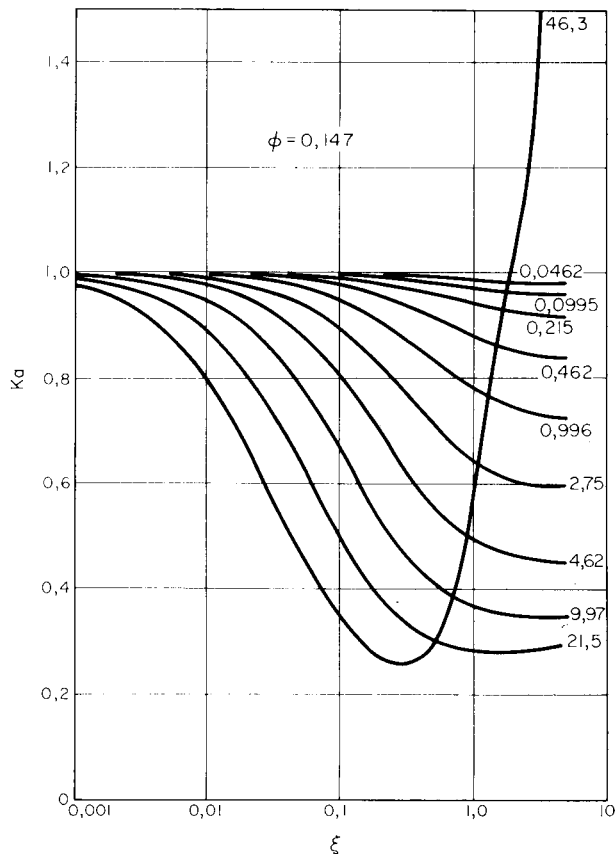


FIG. 15. Function $k_a(\xi)$ versus parameter a for $\phi = 0.147$.

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